# Statistics

Kesondra Key

2026-03-29

IMPORTANT: Change "E:/Github/diurnal-divergence-factor/data/" to your own file location in the setup and on line 263

Purpose: Run statistics and generate tables Note: Not all of the generated .csv's were used!

General information about the dataframe like number of sites and number of droughts per category

```
## # A tibble: 2 x 2
##   drought_cat n_periods
##   <chr>           <int>
## 1 LS                431
## 2 MS                 67
```

```
## # A tibble: 2 x 4
##   drought_cat n_sites median_periods_per_site max_periods_per_site
##   <chr>         <int>                   <dbl>                <int>
## 1 LS               40                     9.5                   24
## 2 MS               15                     5                     10
```

```
## [1] 40
```

```
## [1] 49644
```

```
##    sitename              Date                 VPD              ET
##  Length:49644       Min.   :2001-05-01   Min.   :0.3142   Min.   :-4.381e-05
##  Class :character   1st Qu.:2010-07-11   1st Qu.:1.2126   1st Qu.: 1.471e-05
##  Mode  :character   Median :2015-06-18   Median :1.7479   Median : 4.329e-05
##                     Mean   :2014-11-06   Mean   :2.0286   Mean   : 4.844e-05
##                     3rd Qu.:2019-08-04   3rd Qu.:2.7002   3rd Qu.: 7.654e-05
##                     Max.   :2024-10-31   Max.   :6.6664   Max.   : 2.443e-04
##
##     raw_SWC            SWC              NEE                SW_IN
##  Min.   :-2.225   Min.   :-2.225   Min.   :-29.9636   Min.   :259.9
##  1st Qu.: 4.860   1st Qu.: 4.860   1st Qu.: -8.0431   1st Qu.:478.4
##  Median : 9.702   Median : 9.702   Median : -2.3351   Median :550.8
##  Mean   :13.650   Mean   :13.650   Mean   : -4.5678   Mean   :552.2
##  3rd Qu.:19.415   3rd Qu.:19.415   3rd Qu.: -0.1128   3rd Qu.:627.8
##  Max.   :68.100   Max.   :68.100   Max.   : 17.8967   Max.   :777.6
##  NA's   :4236     NA's   :4236     NA's   :11133      NA's   :40971
##    P_daily_mm       data_source          Latitude       Longitude
##  Min.   :     0.0   Length:49644       Min.   :27.38   Min.   :-121.95
```

```
##  1st Qu.:     0.0   Class :character   1st Qu.:31.79   1st Qu.:-110.87
##  Median :     0.0   Mode  :character   Median :35.97   Median :-109.39
##  Mean   :   150.9                      Mean   :36.57   Mean   :-101.71
##  3rd Qu.:     0.0                      3rd Qu.:39.10   3rd Qu.: -85.31
##  Max.   :121294.1                      Max.   :45.82   Max.   : -76.56
##
##      Veg               mean_AI           Elev          Terrain
##  Length:49644       Min.   :0.1461   Min.   :   5.0   Length:49644
##  Class :character   1st Qu.:0.2354   1st Qu.: 177.0   Class :character
##  Mode  :character   Median :0.5392   Median : 275.0   Mode  :character
##                     Mean   :0.6501   Mean   : 607.6
##                     3rd Qu.:0.9573   3rd Qu.:1188.0
##                     Max.   :2.2119   Max.   :1767.0
##
##    tz_name           TimeZoneHour        Year          Month
##  Length:49644       Min.   :-8.00   Min.   :2001   Min.   : 5.000
##  Class :character   1st Qu.:-7.00   1st Qu.:2010   1st Qu.: 6.000
##  Mode  :character   Median :-7.00   Median :2015   Median : 7.000
##                     Mean   :-6.46   Mean   :2014   Mean   : 7.339
##                     3rd Qu.:-5.00   3rd Qu.:2019   3rd Qu.: 9.000
##                     Max.   :-5.00   Max.   :2024   Max.   :10.000
##
##       Day        two_hr_midday_gap_flag insufficient_cov_day  rain_day
##  Min.   : 1.00   Mode :logical          Mode :logical        Mode :logical
##  1st Qu.: 8.00   FALSE:49644            FALSE:49644          FALSE:41752
##  Median :16.00                                               TRUE :6806
##  Mean   :15.67                                               NA's :1086
##  3rd Qu.:23.00
##  Max.   :31.00
##
##  vpd_flat_day        Sopen         SW_IN_use     Sunny_day         n_total
##  Mode :logical   Min.   :414.3   Min.   :253.4   Mode:logical   Min.   :12.0
##  FALSE:49644     1st Qu.:688.4   1st Qu.:491.4   TRUE:49644     1st Qu.:24.0
##                  Median :729.6   Median :572.3                  Median :24.0
##                  Mean   :730.9   Mean   :568.3                  Mean   :23.5
##                  3rd Qu.:812.8   3rd Qu.:651.9                  3rd Qu.:24.0
##                  Max.   :902.0   Max.   :797.4                  Max.   :24.0
##
##      n_vpd       frac_low_vpd     low_at_6am      low_before_noon
##  Min.   : 7.0   Min.   :0.00000   Mode :logical   Mode :logical
##  1st Qu.:24.0   1st Qu.:0.00000   FALSE:27201     FALSE:26430
##  Median :24.0   Median :0.00000   TRUE :22443     TRUE :23214
##  Mean   :23.4   Mean   :0.07811
##  3rd Qu.:24.0   3rd Qu.:0.13636
##  Max.   :24.0   Max.   :0.47826
##
##  low_vpd_day       SWC_filled     SWC_gapfilled_flag    SWC_pref
##  Mode :logical   Min.   :-2.225   Mode :logical      Min.   :-2.225
##  FALSE:49644     1st Qu.: 4.860   FALSE:49644        1st Qu.: 4.860
##                  Median : 9.702                      Median : 9.702
##                  Mean   :13.650                      Mean   :13.650
##                  3rd Qu.:19.415                      3rd Qu.:19.415
##                  Max.   :68.100                      Max.   :68.100
##                  NA's   :4236                        NA's   :4236
```

```
##   SWC_pref_3d     scaled_raw_SWC       AI_cat             ET_Hour
##  Min.   :-1.469   Min.   :-3.3090   Length:49644       Min.   : 6.00
##  1st Qu.: 4.903   1st Qu.:-0.7984   Class :character   1st Qu.:11.00
##  Median : 9.714   Median :-0.3369   Mode  :character   Median :12.00
##  Mean   :13.666   Mean   :-0.1296                      Mean   :11.96
##  3rd Qu.:19.409   3rd Qu.: 0.4482                      3rd Qu.:13.00
##  Max.   :67.144   Max.   : 6.5729                      Max.   :17.50
##  NA's   :4219     NA's   :4219
##     VPD_Hour           DDF              Cluster        DroughtNo
##  Min.   : 6.00   Min.   :-11.000   Min.   : 1.0    Min.   : 1.000
##  1st Qu.:14.50   1st Qu.: -4.500   1st Qu.: 6.0    1st Qu.: 5.000
##  Median :15.00   Median : -3.000   Median :12.0    Median : 9.000
##  Mean   :15.08   Mean   : -3.122   Mean   :13.8    Mean   : 9.889
##  3rd Qu.:16.00   3rd Qu.: -1.500   3rd Qu.:19.0    3rd Qu.:14.000
##  Max.   :17.50   Max.   :  7.000   Max.   :45.0    Max.   :28.000
##                                    NA's   :38687   NA's   :38687
##  drought_cat        min_scaled_SWC    ClusterStart          Drought
##  Length:49644       Min.   :-3.301   Min.   :2001-07-11   Length:49644
##  Class :character   1st Qu.:-1.533   1st Qu.:2010-07-23   Class :character
##  Mode  :character   Median :-1.245   Median :2014-05-20   Mode  :character
##                     Mean   :-1.336   Mean   :2014-10-01
##                     3rd Qu.:-1.060   3rd Qu.:2019-09-07
##                     Max.   :-0.818   Max.   :2024-10-13
##                     NA's   :38687    NA's   :38687
##       GPP             SW_IN_f
##  Min.   :-11.754   Min.   :253.4
##  1st Qu.:  1.178   1st Qu.:495.1
##  Median :  4.845   Median :577.5
##  Mean   :  7.847   Mean   :571.7
##  3rd Qu.: 13.424   3rd Qu.:656.6
##  Max.   : 39.964   Max.   :797.4
##  NA's   :13923     NA's   :8673

## [1] 415

## [1] 498

## 40 US flux tower sites were selected, covering 498 unique drought periods across 415 site-years.

## # A tibble: 3 x 2
##   drought_cat n_sites
##   <chr>         <int>
## 1 LS               40
## 2 ND               40
## 3 MS               15

## # A tibble: 2 x 2
##   drought_cat n_periods
##   <chr>           <int>
## 1 LS                431
## 2 MS                 67

## # A tibble: 2 x 2
```

```
##   drought_cat n_periods
##   <chr>           <int>
## 1 LS                431
## 2 MS                 67

## # A tibble: 2 x 2
##   AI_cat n_sites
##   <chr>    <int>
## 1 Humid       27
## 2 Arid        13

## # A tibble: 4 x 2
##   Veg   n_sites
##   <chr>   <int>
## 1 GRA        14
## 2 SHR        11
## 3 ENF         8
## 4 DBF         7

## # A tibble: 6 x 3
##   AI_cat drought_cat n_sites
##   <chr>  <chr>         <int>
## 1 Arid   LS               13
## 2 Arid   MS                1
## 3 Arid   ND               13
## 4 Humid  LS               27
## 5 Humid  MS               14
## 6 Humid  ND               27

## # A tibble: 12 x 3
##    Veg   drought_cat n_sites
##    <chr> <chr>         <int>
##  1 DBF   LS                7
##  2 DBF   MS                6
##  3 DBF   ND                7
##  4 ENF   LS                8
##  5 ENF   MS                2
##  6 ENF   ND                8
##  7 GRA   LS               14
##  8 GRA   MS                4
##  9 GRA   ND               14
## 10 SHR   LS               11
## 11 SHR   MS                3
## 12 SHR   ND               11
```

```
## # A tibble: 40 x 8
##    sitename Latitude Longitude Veg   mean_AI AI_cat Terrain          data_source
##    <chr>       <dbl>     <dbl> <chr>   <dbl> <chr>  <chr>            <chr>
##  1 US-AUD       31.6    -111.  GRA     0.235 Arid   <NA>             Ameriflux
##  2 US-CHR       35.9     -84.3 DBF     1.18  Humid  <NA>             Ameriflux
##  3 US-DK2       36.0     -79.1 DBF     0.992 Humid  GENTLE SLOPE (<~ Ameriflux
##  4 US-DK3       36.0     -79.1 ENF     0.994 Humid  GENTLE SLOPE (<~ Ameriflux
##  5 US-IB2       41.8     -88.2 GRA     0.941 Humid  <NA>             Ameriflux
##  6 US-JO1       32.6    -107.  SHR     0.146 Arid   MEDIUM SLOPE (>~ Fluxnet
##  7 US-JO2       32.6    -107.  SHR     0.154 Arid   MEDIUM SLOPE (>~ Ameriflux
##  8 US-KM2       42.4     -85.3 GRA     0.950 Humid  UNDULATED/VARIA~ Ameriflux
##  9 US-KM3       42.4     -85.3 GRA     0.949 Humid  MEDIUM SLOPE (>~ Ameriflux
## 10 US-KON       39.1     -96.6 GRA     0.664 Humid  UNDULATED/VARIA~ Fluxnet
## # i 30 more rows
```

## The Tables1 Section

This runs ANOVA, t-test, and description statistics, generate the information shown within the results, captions, and for tables 2, 3, 4, and S2:

## Table 2

### Descriptive statistics for DDF (hours) during drought and ND periods. Welch's t-test: t(17109.79) = -9.02, p < 0.001; [mean] = 0.21 h, ~13 min

filename: desc_Drought_DDF.csv; ttest_DDF.csv

## Table 3

### Descriptive statistics for DDF (hours) for each drought severity category (ND, LS, MS)

filename: desc_drought_cat_DDF.csv

## Table 4

### Descriptive statistics for DDF (hours) for each aridity category comparing each drought category (e.g., ND, drought)

filename: desc_AI_cat_DDF.csv

## Table S2.

### Summary statistical results for each vegetation (Veg) category

filename: desc_Veg_DDF.csv

## A tibble: 2 x 10

Category group_var test t df p sig baseline comparison
1 Drought (Dr~ Drought t-te~ -9.02 17110. 2.10e- 19 *** Drought ND
2 Aridity cla~ AI_cat t-te~ -35.0 46984. 7.02e-265 *** Arid Humid
# i 1 more variable: med_diff # A tibble: 2 x 9 Category group_var test F df1 df2 p sig omega_sq 1 Drought severity drought_cat ANOVA 117. 2 49641 2.11e-51 *** 0.00465 2 Vegetation type Veg ANOVA 575. 3 49640 0 *** 0.0335

Table 1: Descriptive stats by Drought (response = DDF). CI = 95% t-based CI of the mean.

| Category | Group | sites_n | n_obs | mean | sd | median | ci_low | ci_high |
|---|---|---|---|---|---|---|---|---|
| Drought (Drought vs ND) | Drought | 40 | 10957 | -3.287 | 2.183 | -3.5 | -3.327 | -3.246 |
| Drought (Drought vs ND) | ND | 40 | 38687 | -3.075 | 2.099 | -3.0 | -3.096 | -3.054 |

Table 2: Descriptive stats by drought_cat (response = DDF). CI = 95% t-based CI of the mean.

| Category | Group | sites_n | n_obs | mean | sd | median | ci_low | ci_high | tukey |
|---|---|---|---|---|---|---|---|---|---|
| Drought severity | LS | 40 | 8954 | -3.403 | 2.212 | -3.5 | -3.449 | -3.357 | c |
| Drought severity | MS | 15 | 2003 | -2.766 | 1.966 | -3.0 | -2.852 | -2.680 | a |
| Drought severity | ND | 40 | 38687 | -3.075 | 2.099 | -3.0 | -3.096 | -3.054 | b |

Table 3: Descriptive stats by Veg (response = DDF). CI = 95% t-based CI of the mean.

| Category | Group | sites_n | n_obs | mean | sd | median | ci_low | ci_high | tukey |
|---|---|---|---|---|---|---|---|---|---|
| Vegetation type | DBF | 7 | 8423 | -2.604 | 1.950 | -2.5 | -2.646 | -2.563 | c |
| Vegetation type | ENF | 8 | 7812 | -3.010 | 2.150 | -3.0 | -3.057 | -2.962 | a |
| Vegetation type | GRA | 14 | 17792 | -2.942 | 2.004 | -3.0 | -2.972 | -2.913 | a |
| Vegetation type | SHR | 11 | 15617 | -3.662 | 2.204 | -3.5 | -3.697 | -3.627 | b |

Table 4: Descriptive stats by AI_cat (response = DDF). CI = 95% t-based CI of the mean.

| Category | Group | sites_n | n_obs | mean | sd | median | ci_low | ci_high |
|---|---|---|---|---|---|---|---|---|
| Aridity class | Arid | 13 | 23228 | -3.475 | 2.213 | -3.5 | -3.503 | -3.447 |
| Aridity class | Humid | 27 | 26416 | -2.811 | 1.982 | -3.0 | -2.835 | -2.788 |

Table 5: t-tests

| Category | group_var | test | t | df | p | sig | baseline | comparison | med_diff |
|---|---|---|---|---|---|---|---|---|---|
| Drought (Drought vs ND) | Drought | t-test | -9.0187 | 17109.79 | 0 | *** | Drought | ND | 0.5 |
| Aridity class | AI_cat | t-test | -34.9946 | 46984.06 | 0 | *** | Arid | Humid | 0.5 |

Table 6: ANOVAs + omega squared

| Category | group_var | test | F | df1 | df2 | p | sig | omega_sq |
|---|---|---|---|---|---|---|---|---|
| Drought severity | drought_cat | ANOVA | 116.9620 | 2 | 49641 | 0 | *** | 0.0047 |

| Category | group_var | test | F | df1 | df2 | p | sig | omega_sq |
|---|---|---|---|---|---|---|---|---|
| Vegetation type | Veg | ANOVA | 574.7901 | 3 | 49640 | 0 | *** | 0.0335 |

## The Tables2 Section

Drought severity - this runs ANOVA, t-test, and description statistics, generate the information shown within the results, captions, and for tables 4, 5, S3, and S4:

## Table 4.

### Descriptive statistics for DDF (hours) for each aridity category comparing each drought category (e.g., ND, drought)

filename: desc_AI_cat_x_drought_DDF.csv

## Table 5.

### Descriptive statistics for DDF (hours) for each aridity category comparing each drought severity category

filename: desc_AI_cat_x_drought_cat_DDF.csv

## Table S3.

### Summary statistic results for each vegetation (Veg) type under non-drought (ND) and drought conditions

filename: desc_Veg_x_drought_DDF.csv

## Table S4.

### Summary statistic results for each vegetation (Veg) type under non-drought (ND), less severe (LS) drought, and more severe (MS) drought conditions

filename: desc_Veg_x_drought_cat_DDF.csv

Table 7: Veg × drought (response=DDF); CI = 95% t-based CI of mean

| Category | Veg | Drought | sites_n | n_obs | mean | sd | median | ci_low | ci_high |
|---|---|---|---|---|---|---|---|---|---|
| Veg | DBF | ND | 7 | 6366 | -2.533 | 1.947 | -2.5 | -2.581 | -2.485 |
| Veg | DBF | Drought | 7 | 2057 | -2.825 | 1.942 | -3.0 | -2.909 | -2.741 |
| Veg | ENF | ND | 8 | 6123 | -2.939 | 2.152 | -3.0 | -2.993 | -2.885 |
| Veg | ENF | Drought | 8 | 1689 | -3.268 | 2.124 | -3.5 | -3.369 | -3.166 |
| Veg | GRA | ND | 14 | 13772 | -2.920 | 1.977 | -3.0 | -2.953 | -2.887 |
| Veg | GRA | Drought | 14 | 4020 | -3.018 | 2.095 | -3.0 | -3.082 | -2.953 |
| Veg | SHR | ND | 11 | 12426 | -3.592 | 2.169 | -3.5 | -3.631 | -3.554 |
| Veg | SHR | Drought | 11 | 3191 | -3.933 | 2.317 | -4.0 | -4.013 | -3.852 |

Table 8: t-tests (within each Veg): ND vs Drought (no p adjustment)

| Category | Veg | t | df | p | p_adj | sig | baseline | comparison | med_diff |
|---|---|---|---|---|---|---|---|---|---|
| Veg | DBF | 5.9333 | 3489.361 | 0.0000 | 0.0000 | *** | ND | Drought | -0.5 |
| Veg | ENF | 5.6184 | 2719.256 | 0.0000 | 0.0000 | *** | ND | Drought | -0.5 |
| Veg | GRA | 2.6255 | 6254.608 | 0.0087 | 0.0087 | ** | ND | Drought | 0.0 |
| Veg | SHR | 7.4970 | 4725.072 | 0.0000 | 0.0000 | *** | ND | Drought | -0.5 |

Table 9: Veg × drought_cat (response=DDF); CI = 95% t-based CI of mean

| Category | Veg | drought_cat | sites_n | n_obs | mean | sd | median | ci_low | ci_high |
|---|---|---|---|---|---|---|---|---|---|
| Veg | DBF | LS | 7 | 1137 | -2.834 | 1.879 | -3.0 | -2.943 | -2.724 |
| Veg | DBF | MS | 6 | 920 | -2.815 | 2.019 | -3.0 | -2.945 | -2.684 |
| Veg | DBF | ND | 7 | 6366 | -2.533 | 1.947 | -2.5 | -2.581 | -2.485 |
| Veg | ENF | LS | 8 | 1487 | -3.310 | 2.097 | -3.5 | -3.417 | -3.204 |
| Veg | ENF | MS | 2 | 202 | -2.953 | 2.295 | -3.0 | -3.271 | -2.635 |
| Veg | ENF | ND | 8 | 6123 | -2.939 | 2.152 | -3.0 | -2.993 | -2.885 |
| Veg | GRA | LS | 14 | 3280 | -3.137 | 2.162 | -3.0 | -3.211 | -3.063 |
| Veg | GRA | MS | 4 | 740 | -2.487 | 1.674 | -2.5 | -2.608 | -2.366 |
| Veg | GRA | ND | 14 | 13772 | -2.920 | 1.977 | -3.0 | -2.953 | -2.887 |
| Veg | SHR | LS | 11 | 3050 | -3.946 | 2.321 | -4.0 | -4.029 | -3.864 |
| Veg | SHR | MS | 3 | 141 | -3.642 | 2.224 | -4.0 | -4.012 | -3.271 |
| Veg | SHR | ND | 11 | 12426 | -3.592 | 2.169 | -3.5 | -3.631 | -3.554 |

Table 10: ANOVAs (within each Veg): drought_cat (no p adjustment)

| Category | Veg | F | df1 | df2 | p | sig | omega_sq |
|---|---|---|---|---|---|---|---|
| Veg | DBF | 17.5792 | 2 | 8420 | 0 | *** | 0.0039 |
| Veg | ENF | 18.0245 | 2 | 7809 | 0 | *** | 0.0043 |
| Veg | GRA | 35.5763 | 2 | 17789 | 0 | *** | 0.0039 |
| Veg | SHR | 31.6844 | 2 | 15614 | 0 | *** | 0.0039 |

Table 11: Pairwise t-tests within each Veg: drought_cat vs ND (no p adjustment)

| Category | Veg | drought_cat | t | df | p | p_adj | sig | baseline | med_diff |
|---|---|---|---|---|---|---|---|---|---|
| Veg | DBF | LS | -4.9475 | 1602.9633 | 0.0000 | 0.0000 | *** | ND | -0.5 |
| Veg | DBF | MS | -3.9769 | 1179.7094 | 0.0001 | 0.0001 | *** | ND | -0.5 |
| Veg | ENF | LS | -6.0990 | 2306.7793 | 0.0000 | 0.0000 | *** | ND | -0.5 |
| Veg | ENF | MS | -0.0873 | 212.8303 | 0.9305 | 0.9305 | ns | ND | 0.0 |
| Veg | GRA | LS | -5.2521 | 4670.8438 | 0.0000 | 0.0000 | *** | ND | 0.0 |
| Veg | GRA | MS | 6.7896 | 853.6523 | 0.0000 | 0.0000 | *** | ND | 0.5 |
| Veg | SHR | LS | -7.6403 | 4445.8061 | 0.0000 | 0.0000 | *** | ND | -0.5 |
| Veg | SHR | MS | -0.2624 | 143.0361 | 0.7934 | 0.7934 | ns | ND | -0.5 |

Table 12: Pairwise t-tests within each AI_cat: drought_cat vs ND (no p adjustment)

| Category | AI_cat | drought_cat | t | df | p | p_adj | sig | baseline | med_diff |
|---|---|---|---|---|---|---|---|---|---|
| AI_cat | Arid | LS | -7.0708 | 6448.4237 | 0.0000 | 0.0000 | *** | ND | -0.5 |
| AI_cat | Arid | MS | -2.0577 | 39.1336 | 0.0463 | 0.0463 | * | ND | -1.0 |
| AI_cat | Humid | LS | -10.7944 | 6505.7730 | 0.0000 | 0.0000 | *** | ND | -0.5 |
| AI_cat | Humid | MS | 0.3382 | 2375.2236 | 0.7353 | 0.7353 | ns | ND | -0.5 |

Table 13: Tukey HSD pairwise comparisons within each Veg: drought_cat

| Category | Veg | comparison | group1 | group2 | diff | lwr | upr | p_adj | sig |
|---|---|---|---|---|---|---|---|---|---|
| Veg | DBF | LS-ND | LS | ND | -0.3010 | -0.4479 | -0.1541 | 0.0000 | *** |
| Veg | DBF | MS-ND | MS | ND | -0.2819 | -0.4428 | -0.1210 | 0.0001 | *** |
| Veg | DBF | MS-LS | MS | LS | 0.0191 | -0.1832 | 0.2214 | 0.9734 | ns |
| Veg | ENF | LS-ND | LS | ND | -0.3717 | -0.5171 | -0.2263 | 0.0000 | *** |
| Veg | ENF | MS-ND | MS | ND | -0.0143 | -0.3740 | 0.3454 | 0.9952 | ns |
| Veg | ENF | MS-LS | MS | LS | 0.3574 | -0.0198 | 0.7345 | 0.0677 | ns |
| Veg | GRA | LS-ND | LS | ND | -0.2171 | -0.3082 | -0.1260 | 0.0000 | *** |
| Veg | GRA | MS-ND | MS | ND | 0.4331 | 0.2562 | 0.6101 | 0.0000 | *** |
| Veg | GRA | MS-LS | MS | LS | 0.6502 | 0.4594 | 0.8410 | 0.0000 | *** |
| Veg | SHR | LS-ND | LS | ND | -0.3538 | -0.4580 | -0.2496 | 0.0000 | *** |
| Veg | SHR | MS-ND | MS | ND | -0.0494 | -0.4861 | 0.3872 | 0.9620 | ns |
| Veg | SHR | MS-LS | MS | LS | 0.3044 | -0.1397 | 0.7485 | 0.2429 | ns |

Table 14: Tukey HSD pairwise comparisons within each AI_cat: drought_cat

| Category | AI_cat | comparison | group1 | group2 | diff | lwr | upr | p_adj | sig |
|---|---|---|---|---|---|---|---|---|---|
| AI_cat | Arid | LS-ND | LS | ND | -0.2727 | -0.3589 | -0.1866 | 0.0000 | *** |
| AI_cat | Arid | MS-ND | MS | ND | -0.7915 | -1.6117 | 0.0286 | 0.0613 | ns |
| AI_cat | Arid | MS-LS | MS | LS | -0.5188 | -1.3417 | 0.3041 | 0.3017 | ns |
| AI_cat | Humid | LS-ND | LS | ND | -0.3589 | -0.4356 | -0.2822 | 0.0000 | *** |
| AI_cat | Humid | MS-ND | MS | ND | 0.0156 | -0.0940 | 0.1252 | 0.9407 | ns |
| AI_cat | Humid | MS-LS | MS | LS | 0.3745 | 0.2490 | 0.5000 | 0.0000 | *** |

Table 15: AI_cat × drought (response=DDF); CI = 95% t-based CI of mean

| Category | AI_cat | Drought | sites_n | n_obs | mean | sd | median | ci_low | ci_high |
|---|---|---|---|---|---|---|---|---|---|
| AI_cat | Arid | ND | 13 | 18699 | -3.421 | 2.174 | -3.5 | -3.452 | -3.390 |
| AI_cat | Arid | Drought | 13 | 4529 | -3.698 | 2.355 | -4.0 | -3.767 | -3.630 |
| AI_cat | Humid | ND | 27 | 19988 | -2.752 | 1.971 | -2.5 | -2.779 | -2.725 |
| AI_cat | Humid | Drought | 27 | 6428 | -2.996 | 2.003 | -3.0 | -3.045 | -2.948 |

Table 16: t-tests (within each AI_cat): ND vs Drought (no p adjustment)

| Category | AI_cat | t | df | p | sig | baseline | comparison | med_diff |
|---|---|---|---|---|---|---|---|---|
| AI_cat | Arid | 7.2138 | 6522.242 | 0 | *** | ND | Drought | -0.5 |
| AI_cat | Humid | 8.5486 | 10720.757 | 0 | *** | ND | Drought | -0.5 |

Table 17: AI_cat × drought_cat (response=DDF); CI = 95% t-based CI of mean

| Category | AI_cat | drought_cat | sites_n | n_obs | mean | sd | median | ci_low | ci_high |
|---|---|---|---|---|---|---|---|---|---|
| AI_cat | Arid | ND | 13 | 18699 | -3.421 | 2.174 | -3.5 | -3.452 | -3.390 |
| AI_cat | Arid | LS | 13 | 4489 | -3.694 | 2.355 | -4.0 | -3.763 | -3.625 |
| AI_cat | Arid | MS | 1 | 40 | -4.213 | 2.431 | -4.5 | -4.990 | -3.435 |
| AI_cat | Humid | ND | 27 | 19988 | -2.752 | 1.971 | -2.5 | -2.779 | -2.725 |
| AI_cat | Humid | LS | 27 | 4465 | -3.111 | 2.017 | -3.0 | -3.170 | -3.052 |
| AI_cat | Humid | MS | 14 | 1963 | -2.736 | 1.945 | -3.0 | -2.822 | -2.650 |

Table 18: ANOVAs (within each AI_cat): drought_cat (no p adjustment)

| Category | AI_cat | F | df1 | df2 | p | sig | omega_sq |
|---|---|---|---|---|---|---|---|
| AI_cat | Arid | 29.7805 | 2 | 23225 | 0 | *** | 0.0025 |
| AI_cat | Humid | 61.6672 | 2 | 26413 | 0 | *** | 0.0046 |